Chapter 1 - Mathematical Models

1.1

Write the differential equation for a diffusion problem where the diffusing element decays at a rate proportional to the concentration.

Solution:

The diffusion equation in one dimension is

$$\frac{\partial u}{\partial t} - D\frac{\partial^2 u}{\partial x^2} + \gamma u = f$$

where u is the concentration of the element and f is the source. The model for this problem is

$$\frac{\partial u}{\partial t} - D\frac{\partial^2 u}{\partial x^2} = -\gamma u$$

Writing this out in three dimensions, it instead becomes

$$\frac{\partial u}{\partial t} - D\Delta u = -\gamma u$$

where $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

$\square$

1.3

The temperature distribution in a thin isolated rod is $\frac{T_0}{1+x^2}$ at a specific moment in time. Calculate the heat flux. In what way is it directed?

Solution:

The heat flux is

$$\boldsymbol{j} = -\lambda \boldsymbol{\nabla} T = \frac{\lambda 2 x T_0}{\left(1+x^2\right)^2} \hat{\boldsymbol{x}}$$

The direction is $\hat{\boldsymbol{x}}$

□

1.4

Determine a differential equation for heat conduction in a thin rod, not perfectly isolated. The heat loss at the point x is proportional to the difference between the rod temperature and its surrounding temperature.

Solution:

$$\frac{1}{a}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = -\frac{c}{\lambda}(u - T_0)$$

□

1.5

Determine a differential equation for transversal vertical oscillations of a long horizontal string

(a) under the influence of gravity,

(b) if the surrounding medium additionally exerts a resistance proportional to the velocity.

Solution:

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = f$$

(a) Here, f is simply the gravitational force $F = mg$

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = -\frac{mg}{\rho_l}$$

where ρ_l is the length density.

(b) Add a proportionality factor of $\frac{\partial u}{\partial x}$ which is the velocity

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = -\frac{mg}{\rho_l} - a\frac{\partial u}{\partial x}$$

□

1.8

Let $u(x, y)$ be the temperature in a plate Ω.

(a) How do you formulate a mathematical heat conduction problem with homogenous Neumann boundary conditions?

(b) What physical phenomenon gives rise to homogenous Neumann boundary conditions? Motivate.

In the figures below, the solutions to the problem is shown. Along some boundaries there are homogenous Neumann conditions, and along some there are homogenous Dirichlet conditions.

(c) Left figure shows the vector field, $\boldsymbol{\nabla} u$. Along which parts of the boundary is there homogenous Neumann conditions?

(d) Right figure shows the isotherms ($u =$ constant). Along which parts of the boundary is there homogenous Neumann conditions?

Solution:

(a)

$$\frac{1}{a}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f$$
$$\frac{\partial u}{\partial x} = 0, \text{ on some boundary}$$
$$\frac{\partial u}{\partial y} = 0, \text{ on some boundary}$$

(b) Heat flux on boundary

(c) Top, top right, bottom left.

(d) Left, bottom and top right. □

1.10

Solution:

□

1.11

Solution:

□

1.12

A wide and tall wall with thickness L starts with temperature T_0, same as surroundings. At time $t = 0$ the temperature on one side changes to T_1. At the boundaries on the wall the coefficients are α_0 and α_1. Determine the differential equation, initial values and boundary conditions.

Solution:

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0$$
$$u(L, t) = T_0$$
$$\frac{\partial u(0, t)}{\partial x} = \alpha_0 (u(0, t) - T_1)$$
$$\frac{\partial u(L, t)}{\partial x} = -\alpha_1 (u(L, t) - T_0)$$

□

1.13

Interpret the following in physical meaning for heat conduction.

(a) $u(0, t) = 0$

(b) $\frac{\partial u(0,t)}{\partial x} = 0$

(c) $\frac{\partial u(0,t)}{\partial x} - hu(0, t) = 0$, $\frac{\partial u(L,t)}{\partial x} + hu(L, t) = 0$

Solution:

(a) Temperature at $x = 0$ always 0.

(b) No heat flux at $x = 0$, isolated.

(c) Heat flux on both boundaries, proportional to the difference in temperature between the medium and the surroundings. $\square$

1.14

Solution:

$\square$

1.15

Solution:

$\square$

1.18

Solution:

$\square$

1.21

Solution:

$\square$

1.22

Solution:

□

1.23

Solution:

□

1.26

Solution:

□
